SC4001 – Part B: Chest X-ray Classification

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Generative AI Declaration:

I used **Codex** to assist with the generation of selected code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

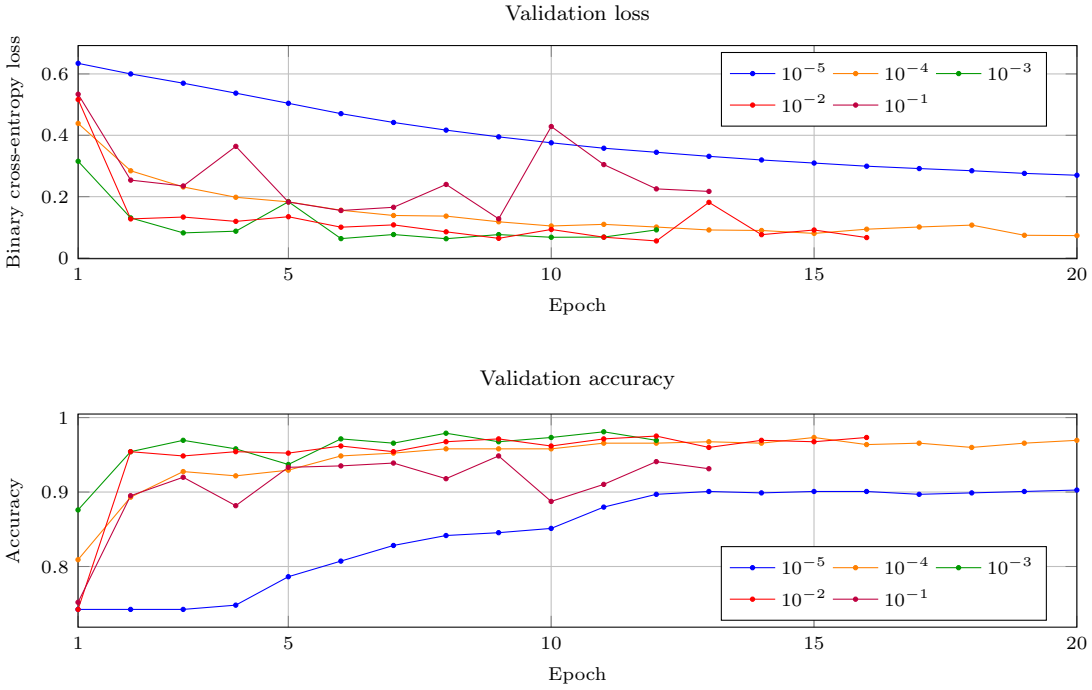
Generative AI Tool Used: Codex

Purpose of Use: Code generation, programming assistance and LaTeX formatting

Verification Method: Manual code review, testing, debugging, and validation against assignment requirements

B1

(a) Validation Loss and Accuracy Curves for Different Learning Rates



(b) Learning Rate Comparison Table

Pneumonia is positive (label 1); the prediction threshold is 0.5. TP, TN, FP, FN denote true positives, true negatives, false positives and false negatives; $N = TP + TN + FP + FN$.

$$\text{Accuracy} = \frac{TP + TN}{N}, \qquad \text{Error} = 1 - \text{Accuracy},$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}.$$

Validation loss is binary cross-entropy, with true label y_i and predicted pneumonia probability p_i:

$$\mathcal{L}_{\text{val}} = -\frac{1}{N} \sum_{i=1}^{N} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

Each row reports the minimum-error checkpoint (ties: lower loss, then earlier epoch). Error, sensitivity and specificity are percentages.

Learning rate	Best epoch	Epochs run	Val. loss	Error (%)	Sens. (%)	Spec. (%)
10^{-5}	20	20	0.2698	9.73	96.14	73.33
10^{-4}	15	20	0.0803	2.67	97.69	96.30
10^{-3}	11	12	0.0683	1.91	97.94	98.52
10^{-2}	12	16	0.0557	2.48	98.20	95.56
10^{-1}	9	13	0.1278	5.15	96.92	88.89

(c) Optimal Learning Rate and Discussion

The optimal learning rate is 10^{-3}, which achieves the lowest validation error of 1.91% at epoch 11. It reaches high validation accuracy within a few epochs. Smaller rates converge more slowly, while larger rates show greater fluctuations. Sensitivity and specificity reflect the model's ability to detect pneumonia cases and correctly identify healthy subjects, respectively. The selected checkpoint achieves 97.94% sensitivity and 98.52% specificity on the validation set.

(d) Test Performance

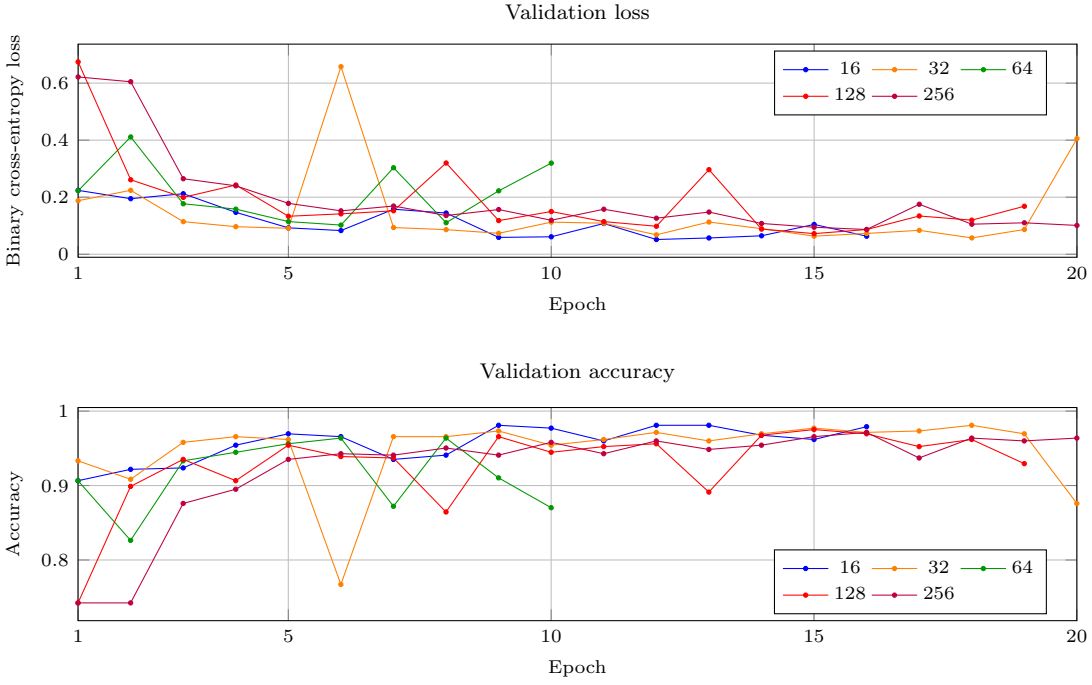
The validation-selected checkpoint (10^{-3}, epoch 11) was evaluated once on 624 test images using training-set normalisation and a threshold of 0.5.

Accuracy (%)	Sensitivity (%)	Specificity (%)
87.02	99.49	66.24

The confusion counts were $TP = 388$, $TN = 155$, $FP = 79$ and $FN = 2$.

B2

(a) Validation Loss and Accuracy Curves for Different Batch Sizes



(b) Batch Size Comparison Table

Metrics follow B1(b). Each row uses the lowest-error checkpoint, with ties resolved by lower validation loss and then earlier epoch.

Batch size	Best epoch	Epochs run	Val. loss	Error (%)	Sens. (%)	Spec. (%)
16	12	16	0.0516	1.91	98.20	97.78
32	18	20	0.0572	1.91	98.97	95.56
64	6	10	0.1022	3.63	97.43	93.33
128	15	19	0.0722	2.48	98.97	93.33
256	16	20	0.0868	2.86	98.20	94.07

(c) Optimal Batch Size and Discussion

Batch sizes 16 and 32 share the lowest validation error of 1.91%. We choose batch size 16 as the optimal one because its validation loss is lower (0.0516 versus 0.0572), and it has a more balanced sensitivity and specificity.

(d) Test Performance

The selected batch-size-16 checkpoint (epoch 12) was evaluated once on 624 test images using training-set normalisation and a threshold of 0.5.

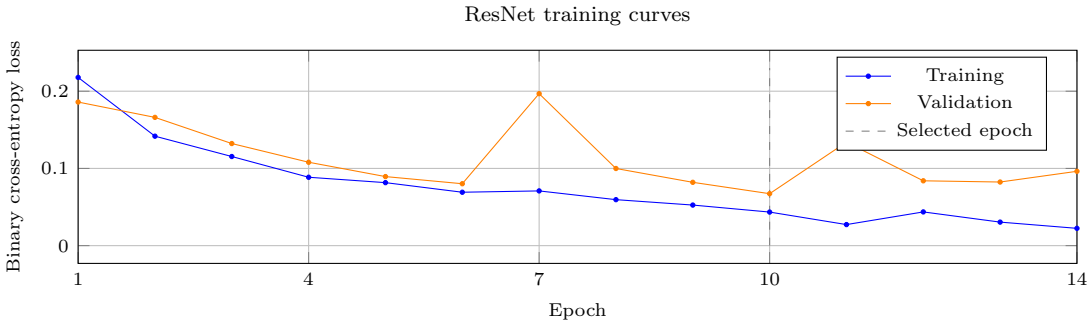
Accuracy (%)	Sensitivity (%)	Specificity (%)
88.62	99.49	70.51

The confusion counts were $TP = 388$, $TN = 165$, $FP = 69$ and $FN = 2$.

B3

(a) Model Performance

The ResNet-like model used learning rate 10^{-3}, batch size 16 and seed 42, with the same preprocessing and early-stopping rules as B1–B2. Training stopped at epoch 14.



The lowest-validation-error checkpoint (epoch 10) was evaluated once on 624 test images with a threshold of 0.5.

Accuracy (%)	Sensitivity (%)	Specificity (%)
87.82	100.00	67.52

The confusion counts were $TP = 390$, $TN = 158$, $FP = 76$ and $FN = 0$.

(b) Comparison and Discussion

Model	Acc. (%)	Sens. (%)	Spec. (%)	Error (%)	Parameters	MFLOPs
AlexNet-like	87.02	99.49	66.24	12.98	99,569	39.06
VGG-like	88.62	99.49	70.51	11.38	18,185	19.46
ResNet-like	87.82	100.00	67.52	12.18	77,169	25.30

Parameters include all trainable weights and biases. FLOPs measure convolutional and fully connected computation per $1 \times 64 \times 64$ image, counting each multiply–accumulate as two operations.

AlexNet-like: Its 5×5 kernels provide larger local receptive fields. The fully connected head contributes to the highest parameter count, and the model has the highest FLOPs and lowest test accuracy and specificity.

VGG-like: Stacked 3×3 convolutions add nonlinear transformations, and global average pooling keeps the classifier compact. It achieves the highest accuracy and specificity with the fewest parameters and FLOPs. Global pooling trades spatial detail for compactness; the model produces 69 false positives.

ResNet-like: Residual shortcuts provide direct gradient paths that can ease optimisation. The model detects all 390 pneumonia cases and produces 76 false positives. VGG offers higher accuracy and specificity at lower computational cost.

Model	Pros	Cons
AlexNet-like	Larger local receptive fields from 5×5 kernels; high sensitivity (99.49%).	Highest parameter count and FLOPs; lowest specificity (66.24%).
VGG-like	Highest accuracy (88.62%) and specificity (70.51%); lowest parameter count and FLOPs.	Global pooling compresses spatial detail; 69 false positives.
ResNet-like	Residual shortcuts support gradient flow; detects all 390 pneumonia cases.	More computation than VGG; 76 false positives and lower specificity (67.52%).

B4

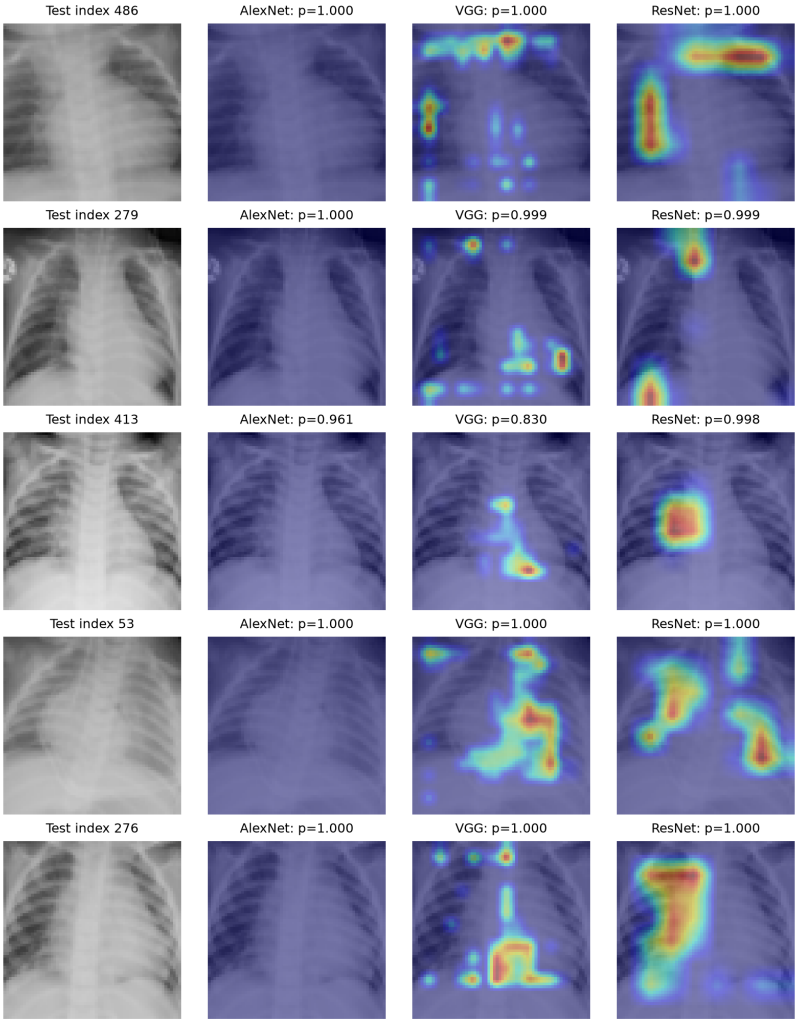
(a) X-ray Selection and Grad-CAM Implementation

Five pneumonia test images (zero-based indices 486, 279, 413, 53 and 276) were sampled without replacement using seed 42. The selected B1–B3 checkpoints used evaluation mode and saved training normalisation. Captum's LayerGradCam targeted the pneumonia logit at the final main-path convolution:

$$\alpha_k = \frac{1}{HW} \sum_{i,j} \frac{\partial z}{\partial A^k_{ij}}, \qquad L_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k A^k \right).$$

Here z is the pneumonia logit and A^k is channel k of the convolutional output. AlexNet and VGG produce 16×16 maps; ResNet produces 8×8 maps. Each map was bilinearly resized to 64×64, min–max normalised and overlaid with opacity 0.4. Constant maps were assigned zero intensity.

(b) Grad-CAM Overlays



Original X-rays and pneumonia Grad-CAM overlays. Red indicates stronger positive attribution; blue indicates low or zero attribution. p is the predicted pneumonia probability. Colours are scaled independently within each map.

(c) Comparison and Interpretation

All three models classify the five images as pneumonia. AlexNet's weighted responses at the selected convolution are entirely negative, so ReLU produces zero maps despite probabilities of 0.961–1.000. This exposes a limitation of positive Grad-CAM at this layer.

VGG produces smaller, scattered hotspots that overlap some brighter lower central and image-right areas in images 53 and 276. ResNet produces broader hotspots with its coarser 8×8 maps. Both highlight upper chest and peripheral regions in image 486. For image 413, VGG emphasises the lower central region and ResNet the mid-chest.